

Threshold Estimation for the Surface Code with the MPS-MPO Decoder

1 Setup

We study the rotated surface code of distance $d \in \{3, 5, 7, 9, 11\}$ under independent, identically distributed bit-flip noise with physical error rate p . Each code encodes one logical qubit in d^2 physical qubits. The decoder is the MPS-MPO contractor with maximum bond dimension $\chi_{\max} = 250$, which approximates maximum-likelihood (ML) decoding by contracting the tensor-network representation of the full posterior distribution over Pauli errors.

For each distance and error rate we run N_{exp} independent decoding trials and record the logical failure rate

$$p_L(p, d) = \frac{\text{number of logical failures}}{N_{\text{exp}}}. \quad (1)$$

The resulting curves are shown in Figure 1.

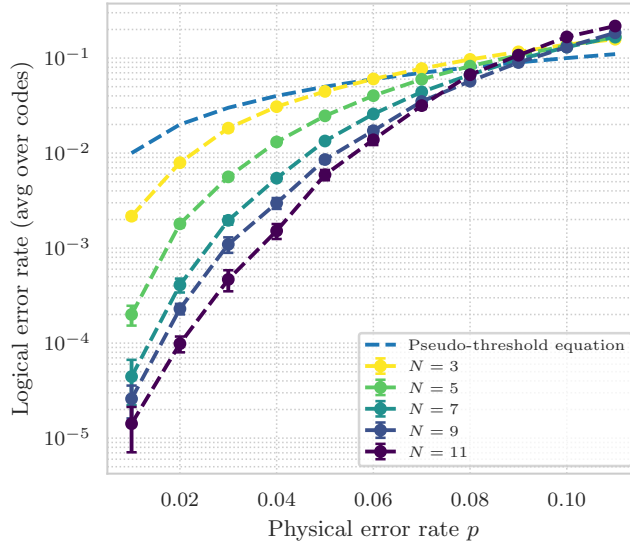


Figure 1: Logical error rate p_L as a function of physical bit-flip rate p for surface codes of distance $d = 3, 5, 7, 9, 11$ decoded with the MPS-MPO decoder ($\chi_{\max} = 250$). The dashed line marks the pseudo-threshold $p_L = p$, i.e. the performance of the trivial (no-correction) decoder.

2 Threshold from Curve Crossings

Below the error-correction threshold p_{th} , larger codes suppress errors faster: p_L decreases as d increases at fixed $p < p_{\text{th}}$. Above the threshold the ordering reverses. The threshold is therefore the value of p at which the logical failure-rate curves for successive distances d and d' intersect:

$$p_{\text{th}}^{(d,d')} = \{p \mid p_L(p, d) = p_L(p, d')\}. \quad (2)$$

In the thermodynamic limit $d \rightarrow \infty$ these crossing points converge to the true threshold. For finite d the crossings depend on the pair (d, d') and provide a family of estimates that can be extrapolated.

In practice we fit a cubic spline to each empirical curve $p_L(p, d)$ and locate the crossing numerically using Brent’s method. The results are summarised in Table 1.

Table 1: Pairwise threshold estimates from the MPS-MPO decoder data.

Distance pair (d, d')	$p_{\text{th}}^{(d, d')}$	Note
(3, 5)	10.44%	
(5, 7)	10.73%	
(7, 9)	10.00%	
(9, 11)	7.30%	bond dim. not converged

The (9, 11) crossing is a clear outlier: at small p the logical failure rate of the $d = 11$ code is not yet converged in χ_{max} , causing the $d = 11$ curve to lie above the $d = 9$ curve and producing a spuriously low crossing. Excluding this pair, the mean of the three reliable estimates gives

$$\hat{p}_{\text{th}} \approx 10.4\%. \quad (3)$$

3 Comparison with Known Values

The surface code under independent bit-flip noise admits two well-known reference thresholds.

Minimum-weight perfect matching (MWPM). The MWPM decoder maps decoding to a minimum-weight perfect matching problem on the syndrome graph. Numerical studies place its threshold at $p_{\text{th}}^{\text{MWPM}} \approx 10.31\%$ [1].

Maximum-likelihood (ML) decoder. The ML threshold for the surface code under bit-flip noise corresponds to the ferromagnetic–paramagnetic phase boundary of the two-dimensional random-bond Ising model (RBIM) evaluated on the Nishimori line [1, 2]. Numerical transfer-matrix studies give

$$p_{\text{th}}^{\text{ML}} \approx 10.94\%. \quad (4)$$

This is the highest threshold achievable by any decoder for this noise model.

MPS-MPO decoder. Our estimate $\hat{p}_{\text{th}} \approx 10.4\%$ lies between the MWPM and ML values, as expected for a finite- χ MPS decoder: larger bond dimension recovers more correlations and pushes the threshold toward the ML limit. The gap to the ML value indicates that $\chi_{\text{max}} = 250$ is not yet sufficient to reach exact ML decoding at the distances considered, consistent with the bond-dimension scaling results in Figure 2 of the main text.

References

- [1] E. Dennis, A. Kitaev, A. Landahl, J. Preskill, *Topological quantum memory*, J. Math. Phys. **43**, 4452 (2002).
- [2] C. Wang, J. Harrington, J. Preskill, *Confinement Higgs transition in a disordered gauge theory and the accuracy threshold for quantum memory*, Ann. Phys. **303**, 31 (2003).